

C - Long Sequence

分值: 300 分

Problem Statement

You are given integers N and K , along with N integer sequences $A_1, A_2, \dots, A_N$ and a length- N integer sequence $C = (C_1, C_2, \dots, C_N)$. The length of integer sequence A_i is L_i , and $A_i = (A_{i,1}, A_{i,2}, \dots, A_{i,L_i})$. It is guaranteed that $1 \leq K \leq \sum_{i=1}^N C_i L_i$.

给定整数 N 和 K , 以及 N 个整数序列 $A_1, A_2, \dots, A_N$ 和一个长度为 N 的整数序列 $C = (C_1, C_2, \dots, C_N)$ 。整数序列 A_i 的长度为 L_i , 且满足 $A_i = (A_{i,1}, A_{i,2}, \dots, A_{i,L_i})$ 。保证 $1 \leq K \leq \sum_{i=1}^N C_i L_i$ 。

Construct an integer sequence $B = (B_1, B_2, \dots, B_{\sum_{i=1}^N C_i L_i})$ from A and C using the following procedure.

通过以下步骤, 由 A 和 C 构造整数序列 $B = (B_1, B_2, \dots, B_{\sum_{i=1}^N C_i L_i})$ 。

- Let B be an integer sequence of length 0.
- For $i = 1, 2, \dots, N$ in this order, perform the following operation:
 - Do this C_i times: append A_i to the end of B .
- 设 B 是一个长度为 0 的整数序列。
- 按顺序对每个 $i = 1, 2, \dots, N$ 执行以下操作:
 - 重复 C_i 次: 将 A_i 追加到 B 的末尾。

Find the value of B_K .

求 B_K 的值。

Constraints

- $1 \leq N$
- $1 \leq L_i$

- $\sum_{i=1}^N L_i \leq 2 \times 10^5$
- $1 \leq A_{i,j} \leq 10^9 \ (1 \leq j \leq L_i)$
- $1 \leq C_i \leq 10^9$
- $1 \leq K \leq \sum_{i=1}^N C_i L_i$
- All input values are integers.
- $1 \leq N$
- $1 \leq L_i$
- $\sum_{i=1}^N L_i \leq 2 \times 10^5$
- $1 \leq A_{i,j} \leq 10^9 \ (1 \leq j \leq L_i)$
- $1 \leq C_i \leq 10^9$
- $1 \leq K \leq \sum_{i=1}^N C_i L_i$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```

N K
L1 A1,1 A1,2 ... A1,L1
L2 A2,1 A2,2 ... A2,L2
⋮
LN AN,1 AN,2 ... AN,LN
C1 C2 ... CN

```

Output

Output the answer.

输出答案。

Sample Input 1

```
3 9
3 1 3 2
1 3
2 4 3
1 3 2
```

Sample Output 1

4

B is constructed as follows:

B 的构造过程如下:

- Let $B = ()$.
- Append A_1 to the end of B once. We get $B = (1, 3, 2)$.
- Append A_2 to the end of B three times. We get $B = (1, 3, 2, 3, 3, 3)$.
- Append A_3 to the end of B twice. We get $B = (1, 3, 2, 3, 3, 3, 4, 3, 4, 3)$.
- 设 $B = ()$.
- 将 A_1 追加到 B 的末尾一次, 得到 $B = (1, 3, 2)$.
- 将 A_2 追加到 B 的末尾三次, 得到 $B = (1, 3, 2, 3, 3, 3)$.
- 将 A_3 追加到 B 的末尾两次, 得到 $B = (1, 3, 2, 3, 3, 3, 4, 3, 4, 3)$.

For $B = (1, 3, 2, 3, 3, 3, 4, 3, 4, 3)$, we have $B_9 = 4$, so output 4.

对于 $B = (1, 3, 2, 3, 3, 3, 4, 3, 4, 3)$, 有 $B_9 = 4$, 因此输出 4.

Sample Input 2

```
3 1
1 7
1 111
1 5
1 100 10000
```

Sample Output 2

7

Sample Input 3

```
3 3163812
5 1 2 3 4 5
4 9 8 7 6
2 10 11
87043 908415 9814
```

Sample Output 3

9